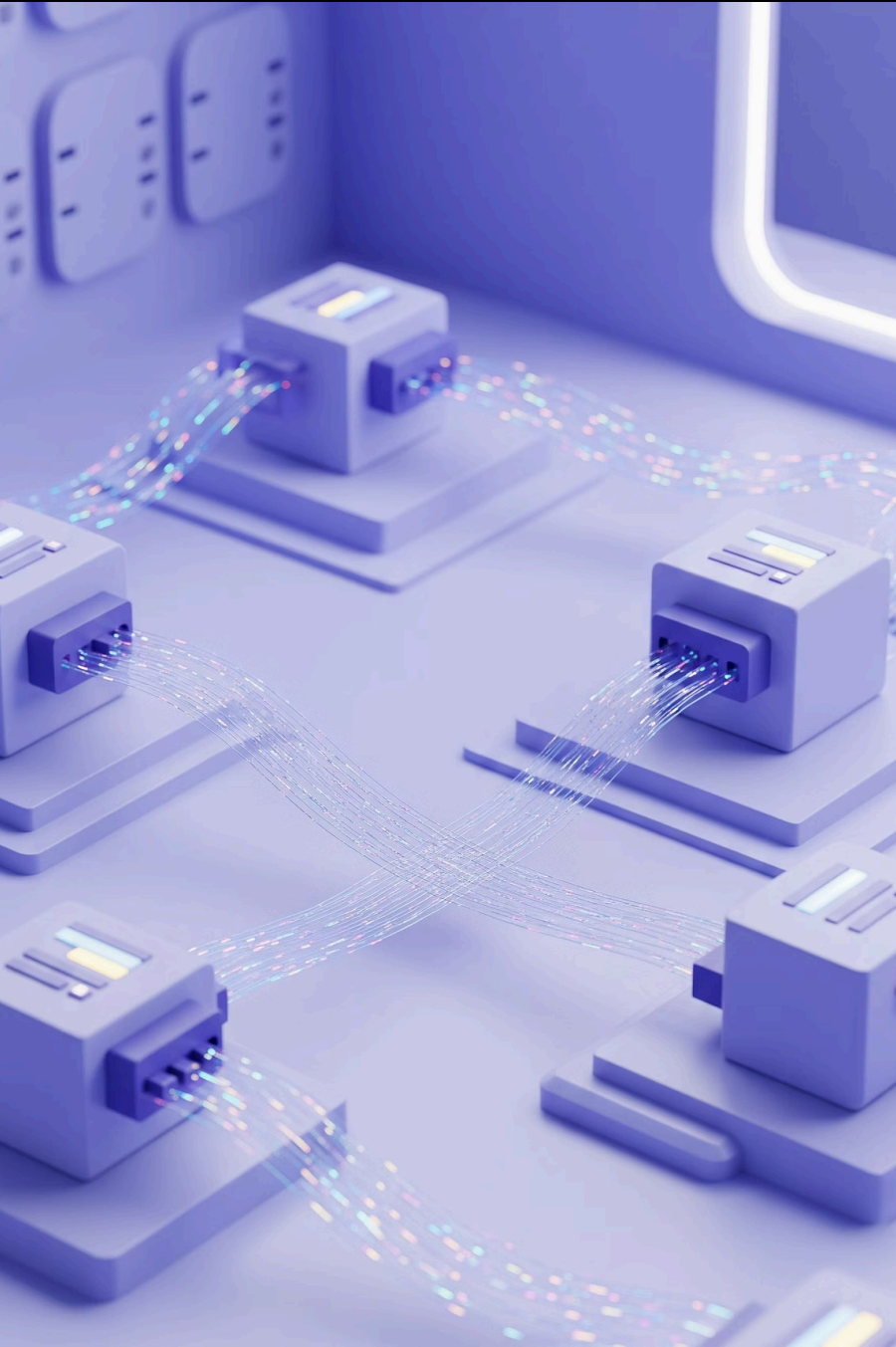




Introduction to ETL and ELT Pipelines

Building Scalable Data Workflows for AI Infrastructure

Modern AI systems require robust data infrastructure. Let's explore how ETL and ELT pipelines form the backbone of effective data workflows that power AI applications.

Why Data Pipelines Matter



Foundation for AI

AI models rely on large volumes of clean, structured data to perform effectively

Automation

Pipelines automate the journey from raw data to usable insights, reducing manual effort and errors

Scalability

Well-designed pipelines ensure reproducibility and can scale with growing data needs

What Is ETL?



Extract

Pull data from multiple sources
(databases, APIs, files)

Handle various formats and
protocols



Transform

Clean, normalize, and structure the
data

Apply business rules and
aggregations



Load

Store processed data in target
systems

Optimize for downstream
consumption

Traditional approach optimized for BI and batch processing workflows

What Is ELT?

Extract

Gather raw data from source systems without preprocessing

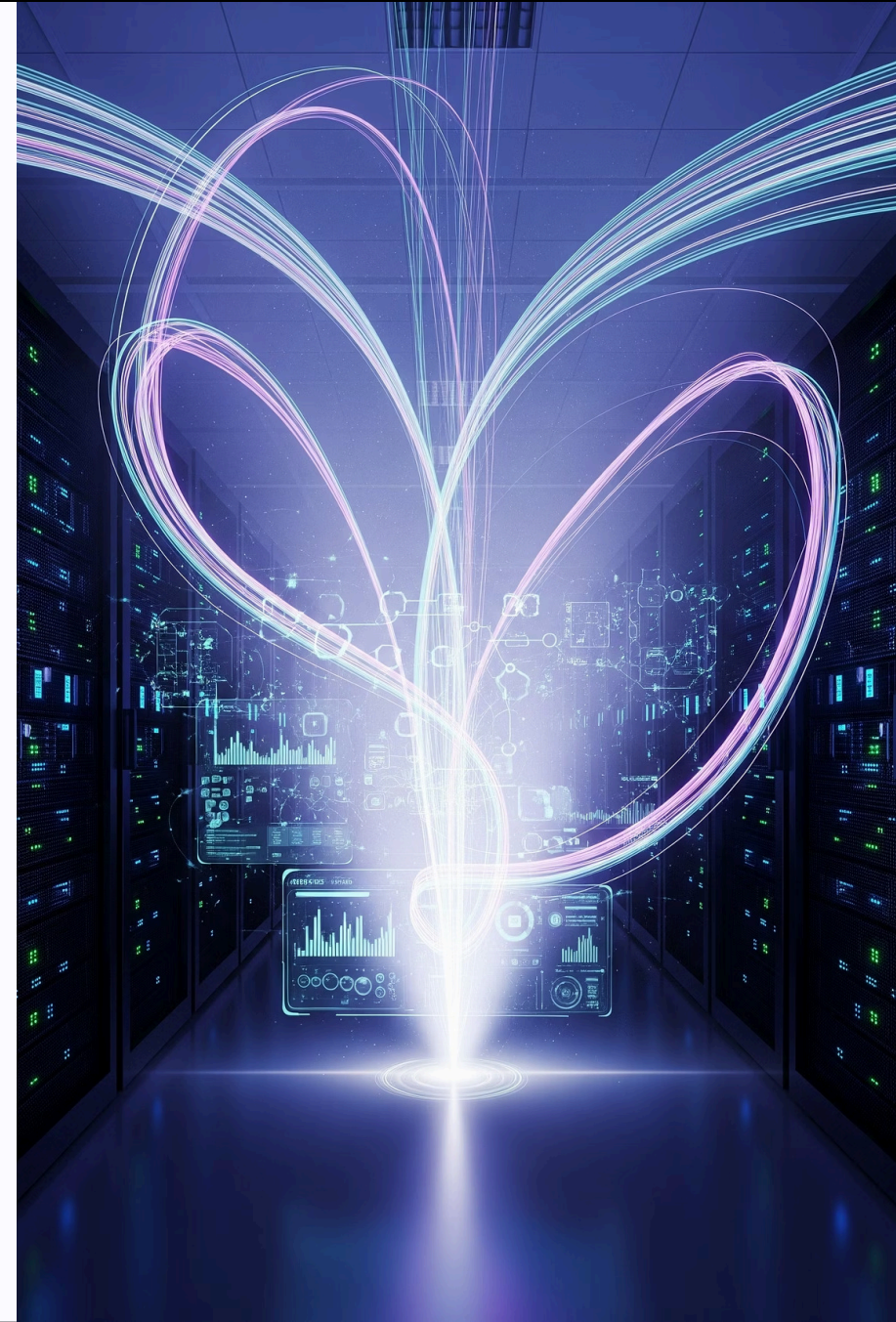
Load

Store raw data directly in a data lake or warehouse

Transform

Leverage in-warehouse compute for processing after loading
Transform data on-demand for specific use cases

Modern approach optimized for cloud-native, large-scale AI workloads



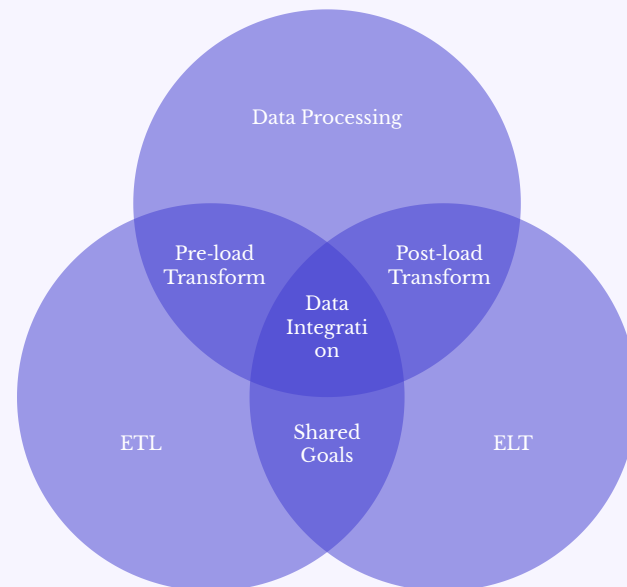
ETL vs ELT: Key Differences

ETL Approach

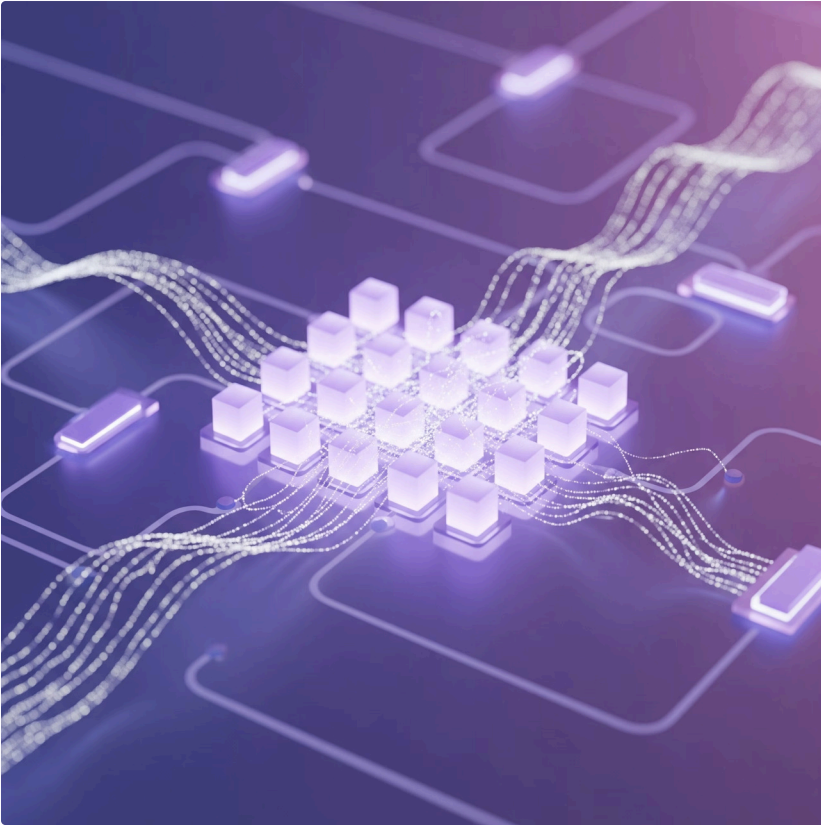
- Process data before loading
- Optimized for smaller datasets
- Well-suited for legacy, on-premises systems
- Transformation happens in a separate processing layer

ELT Approach

- Process data after loading
- Designed for large, raw datasets
- Ideal for modern cloud and AI workflows
- Transformation uses data warehouse compute power



AI Use Cases of Pipelines



1 Data Preprocessing

Scale processing of text, images, audio for AI training

2 Real-time Analytics

Stream IoT sensor data for immediate AI decision-making

3 Feature Engineering

Create and transform model features at scale

4 Knowledge Bases

Feed LLMs with structured, high-quality information

Common Tools in ETL/ELT

ETL Tools

- Talend
- Informatica PowerCenter
- Apache NiFi
- Microsoft SSIS

ELT Tools

- dbt (data build tool)
- Snowflake
- Google BigQuery
- Databricks

Orchestration

- Apache Airflow
- Prefect
- Dagster
- Luigi

Streaming

- Apache Kafka
- Apache Flink
- Apache Spark Streaming
- AWS Kinesis

Each category serves different needs in the data pipeline ecosystem

Best Practices



Error Handling

Implement robust error handling with automated retries and clear logging



Data Quality

Continuously monitor data quality with automated tests and alerting



Modularity

Design for modularity and scalability to handle growing data volumes



Alignment

Align pipeline design with downstream AI/ML needs for optimal performance

Key Takeaways

1

ETL is Traditional

Pre-load transformations

Structured workflows

Mature ecosystem

2

ELT is Modern

Post-load transformations

Flexible processing

Cloud-native design

3

Choose Wisely

Based on workload

Infrastructure constraints

Scale requirements

- ❏ Both approaches are essential components of robust AI data infrastructure. The right choice depends on your specific use case, existing technology stack, and future scalability needs.

Questions? Contact the Data Engineering Team